

Symmetry as a Design Axis in Variational Quantum Learning

Markus Baumann¹, Claudia Linnhoff-Popien¹, and Jonas Stein¹

¹QAR-Lab, Department of Computer Science, LMU Munich, Munich, Germany

Abstract—The success of variational quantum learning models crucially depends on choosing parametrizations that reflect the structure of the problem at hand. Symmetries provide one of the clearest such structures: whenever transformations of the input leave the desired outcome unchanged, this invariance should be built into the model rather than discovered during training. However, imposing a symmetry does not by itself determine a useful ansatz. Even within the symmetry-preserving space, one must decide where the trainable degrees of freedom should be placed. In this work, we study this remaining design freedom in equivariant variational quantum circuits. Building on symmetry-based parameter sharing, we disentangle two architectural choices: how much symmetry should be enforced, and which symmetry-respecting interactions should be trainable. Using Tic-Tac-Toe as a fully enumerable and structurally transparent test case, we find that suitable subgroups preserve most of the generalization benefit. By contrast, the dominant gains arise from gates acting directly on decisive task motifs. Thus, effective equivariant ansätze require two biases: a symmetry bias that restricts the search space, and a task bias that places trainable interactions on the decisive motifs.

Index Terms—quantum machine learning, equivariance, symmetry, inductive bias, variational quantum circuits, parameter sharing, generalization

I. INTRODUCTION

A central challenge in variational quantum machine learning is not merely to build expressive quantum circuits, but to build the right expressive circuits. Layered parameterized circuits with data re-uploading [1], [2] can be highly flexible, but flexibility alone is not a design principle. Without architectural guidance, a model may spend parameters and data on distinctions irrelevant to the task, while failing to allocate capacity to the structures that determine the label. The practical question is therefore not only how much expressivity a quantum model has, but where this expressivity is placed.

Symmetry is one of the few sources of prior knowledge that can be stated exactly. Consider a target map $y : \mathcal{X} \rightarrow \mathcal{Y}$ and a group S acting on the input space through transformations $V_s : \mathcal{X} \rightarrow \mathcal{X}$. The task is invariant under S if

$$y(V_s[x]) = y(x) \quad \text{for all } x \in \mathcal{X}, s \in S.$$

In words, a symmetry specifies transformations of the input that must not change the correct prediction. A translated image still depicts the same object, a permuted graph still represents the same graph, and a rotated Tic-Tac-Toe board still has the same winner. A model that respects this structure by construction need not rediscover it from data.

This observation is central to geometric deep learning: classical architectures such as convolutional neural networks organize trainable degrees of freedom around task symmetries [3]. In quantum machine learning, geometric quantum machine learning represents known transformations on the Hilbert space and constrains circuits accordingly [4], [5]. Related work has developed group-invariant models, equivariant quantum neural networks, permutation-equivariant architectures, graph-equivariant circuits, and symmetry-informed variational ansätze [4]–[7].

Meyer et al. [8] gave a concrete blueprint for this program. If a data embedding $U(x)$ satisfies

$$U(V_s[x]) = U_s U(x) U_s^\dagger,$$

then the data symmetry is represented by a unitary action U_s . Trainable blocks that commute with this representation, together with compatible state preparation and readout, yield a model whose prediction is invariant by architecture rather than by optimization.

However, symmetry alone does not specify an ansatz. It tells us which inputs should be identified, which parameters may be shared, and which observables should be aggregated, but not where trainable interactions should live. Two circuits can respect the same group action and still match the task very differently. We therefore treat symmetry not as a switch, but as a constrained design space in which one must still choose the sharing strength and the motifs on which trainable quantum interactions are placed.

Tic-Tac-Toe is well suited for this question. The task is finite, exactly enumerable, and structurally explicit: the relevant board symmetries, labels, and decisive three-cell motifs are all known in advance. This lets us isolate ansatz-design choices rather than infer them from a large, opaque benchmark.

We separate two choices that are often entangled. The first is how much of the available board symmetry should be enforced: full equivariance is natural, but smaller subgroups may preserve most of the useful bias while leaving the circuit more flexible. The second is which symmetry-respecting interactions should be made trainable. The baseline captures the board’s local geometry, but the target function is decided by complete winning triples. Our extension therefore asks whether placing trainable interactions directly on those motifs improves the equivariant ansatz.

Our results support this design-oriented view. Enforcing symmetry improves the edge baseline, and suitable subgroup sharing retains most of that benefit. The largest gain, however,

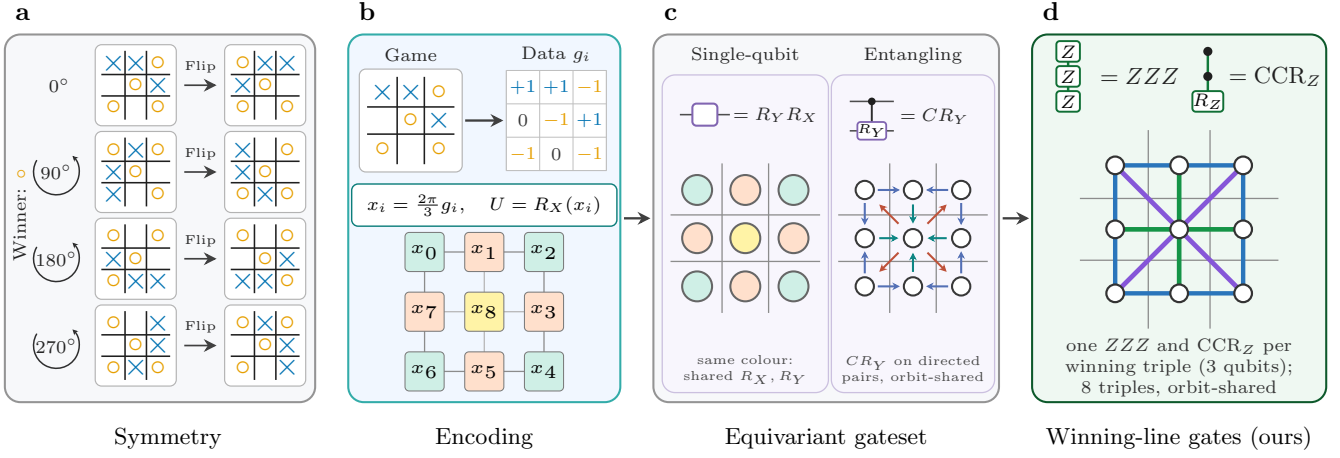


Fig. 1. From task symmetry to a task-aligned equivariant circuit. (a) Tic-Tac-Toe labels are invariant under the D_4 rotations and reflections of the board. (b) Board entries $g_i \in \{-1, 0, +1\}$ are encoded as $R_X(x_i)$ rotations with $x_i = \frac{2\pi}{3}g_i$; the chosen indexing exposes the corner, side, and center orbits. (c) The Meyer et al. [8] edge ansatz uses orbit-shared $R_Y R_X$ rotations and directed $C R_Y$ entanglers on neighbouring and center-connected fields. (d) Our extension adds orbit-shared ZZZ and symmetric CCR_Z gates on the eight winning triples, directly matching the label-relevant motifs while preserving equivariance. Panel layout adapted from Meyer et al. [8].

comes from enriching the equivariant circuit with task-aligned interactions. Thus, symmetry is not merely a constraint that makes a model smaller. It is a language for organizing circuit design, but the effectiveness of the model still depends on placing trainable degrees of freedom where the structure of the task actually lives.

II. PROTOCOL

Each legal, reachable Tic-Tac-Toe board is represented by nine values $g_i \in \{-1, 0, +1\}$ for circle, empty, and cross. The three labels are circle win, draw, and cross win, encoded as $(+1, -1, -1)$, $(-1, +1, -1)$, and $(-1, -1, +1)$. As in Meyer et al. [8], unfinished games are counted as draws. Since the full set contains only 5478 legal, reachable boards, we enumerate it exactly and draw class-balanced train and test splits from this set. The labels are invariant under the eight rotations and reflections of the board, forming the dihedral group D_4 ; this symmetry is illustrated in Fig. 1(a).

We use one qubit per field, as shown in Fig. 1(b). Each value g_i is mapped to the angle $x_i = 2\pi g_i/3$ and encoded by an R_X rotation,

$$U(g) = \bigotimes_{i=0}^8 R_X\left(\frac{2\pi}{3}g_i\right).$$

The fields are numbered cyclically around the rim, with the center last. This makes the three D_4 position orbits explicit: corners $\{0, 2, 4, 6\}$, sides $\{1, 3, 5, 7\}$, and center $\{8\}$.

We separate the imposed symmetry from the choice of gates. Let $H \subseteq D_4$ be the subgroup that we enforce. For $h \in H$, let V_h act on the board and let P_h be the corresponding qubit permutation. The encoding is equivariant: transforming the board before encoding is equivalent to encoding first and then permuting the qubits,

$$U(V_h g) = P_h U(g) P_h^\dagger.$$

A trainable block $W_H(\theta)$ preserves this symmetry if

$$P_h W_H(\theta) P_h^\dagger = W_H(\theta) \quad \text{for all } h \in H.$$

In practice, we enforce this by sharing parameters across gates whose supports lie in the same H -orbit. For directed gates, the order of the support is kept fixed, so control and target roles are not accidentally exchanged. Thus the symmetry fixes the sharing pattern, but not the gate family itself.

The circuit uses repeated data re-uploading layers. Each layer applies the encoding $U(g)$ followed by p repetitions of a trainable block. Unless stated otherwise, we use $L = 3$ layers and $p = 2$ repetitions, keeping depth fixed across all sweeps.

Our baseline is the edge ansatz of Meyer et al., shown in Fig. 1(c). It applies local $R_Y R_X$ rotations and directed controlled- R_Y entanglers on neighbouring rim fields and selected center connections. Symmetry enters only through parameter sharing: gates in the same H -orbit share the same parameter or parameter tuple.

The readout averages Pauli- Z expectations over the three position orbits,

$$r_O(g) = \frac{1}{|O|} \sum_{i \in O} \langle Z_i \rangle_g,$$

for corners, sides, and center. These three values are used as class scores. Since the readout orbits are invariant under all subgroups considered here, the readout is compatible with every symmetry sweep.

This gives our first design axis: the amount of symmetry imposed. We keep encoding, edge wiring, gate family, depth, and readout fixed, and vary only the subgroup H . With r denoting a 90° rotation and f a reflection, we compare $\{e\}$, $\langle r^2 \rangle$, $\langle f \rangle$, $\langle r \rangle = C_4$, $\langle r^2, f \rangle = D_2$, and $\langle r, f \rangle = D_4$. For the edge ansatz at $L = 3, p = 2$, these variants have 204, 108, 126, 60, 78, and 54 trainable parameters, respectively.

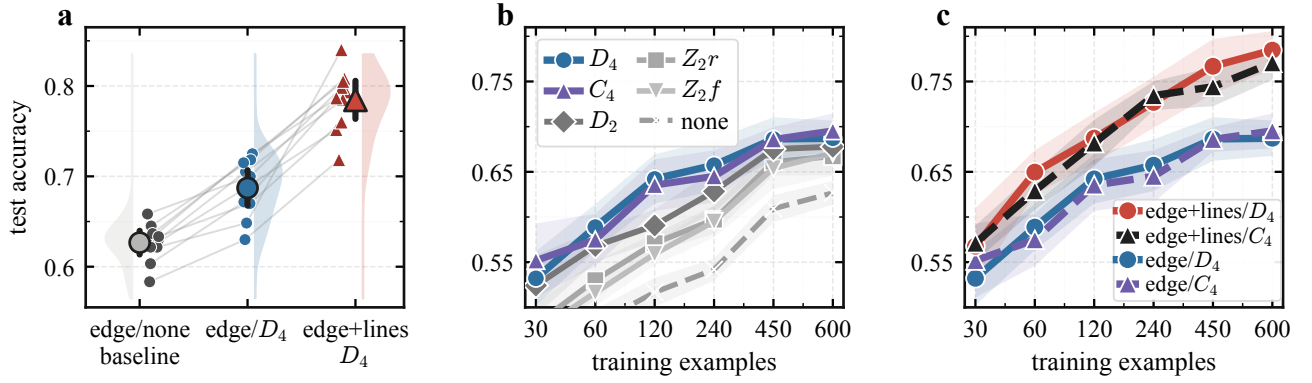


Fig. 2. Main empirical evidence. (a) Full D_4 equivariance improves the original edge ansatz over no sharing. (b) The subgroup sweep across training sizes shows that C_4 tracks D_4 closely. (c) Adding winning-line interactions to the equivariant edge ansatz improves test accuracy while preserving the imposed symmetry.

The second design axis is the interaction structure. The edge ansatz couples neighbouring fields, but Tic-Tac-Toe labels are decided by complete three-cell winning lines. We therefore add orbit-shared gates on the eight winning triples: three rows, three columns, and two diagonals, as shown in Fig. 1(d). Each triple receives one ZZZ rotation and one CCR_Z interaction. The resulting edge+lines model keeps the original edge ansatz but adds trainable interactions directly on the label-relevant motifs. At $L = 3, p = 2$, the full- D_4 edge model has 54 parameters, while the full- D_4 edge+lines model has 90.

III. RESULTS

This section evaluates the two design axes introduced above: the strength of imposed symmetry and the choice of trainable interactions. We first compare no sharing, partial subgroup sharing, and full D_4 sharing with the edge circuit fixed. We then add winning-line interactions and use controls to separate symmetry-aware sharing from mere parameter reduction. All runs use mean-squared error, Adam with learning rate 0.01 [9], batch size 15, 30 minibatch updates per epoch, 100 epochs, fixed seed blocks, PennyLane simulation [10], and PyTorch optimization [11].

Across these experiments, three trends emerge, summarized in Fig. 2. First, the unconstrained edge ansatz has more parameters but generalizes worse than the D_4 -equivariant edge model. At 600 training examples, edge/none reaches about 0.63 test accuracy, while edge/ D_4 reaches about 0.69 with a smaller train-test gap. This supports the basic equivariance effect: the correct board symmetry improves generalization by shaping the hypothesis class.

Second, the symmetry effect is not strictly all-or-nothing. Full D_4 remains the natural symmetry choice, but C_4 rotation sharing tracks it closely across the data grid and is slightly ahead at the smallest training size. We interpret this as evidence that symmetry strength is a tunable architectural bias: partial symmetry can preserve most of the benefit while leaving the circuit less constrained. The random-sharing control in Fig. 3(a) confirms that the gain is not caused by parameter

reduction alone; parameter-matched sharing without group-orbit structure performs worse.

The largest improvement comes from changing the interaction structure. Adding winning-line gates to the equivariant edge circuit, as shown in Fig. 2(c), raises the best D_4 model from about 0.69 to roughly 0.80 test accuracy, with the C_4 variant close behind. The edge ansatz can only represent rows, columns, and diagonals indirectly, whereas the line gates act directly on the motifs that determine the label. This is the central result: symmetry helps, but task-aligned equivariant interactions help more.

The controls in Fig. 3 rule out simpler explanations. The effect is not arbitrary parameter reduction, because random sharing underperforms orbit sharing. It is also not just parameter count: edge+lines/ D_4 has 90 parameters, far fewer than the 204-parameter unshared edge baseline, yet performs better. The ablations show that line channels help on their own, but the strongest model combines edge interactions with both winning-line channels.

IV. DISCUSSION AND OUTLOOK

We have investigated what remains to be designed once a variational quantum learning model has been constrained to respect a data symmetry. Equivariance is a powerful inductive bias, but it is not a complete ansatz prescription. It tells the model which inputs should be treated as equivalent and how parameters should be shared across symmetry-related parts of the data. It leaves open how the model's trainable capacity should be assigned to the structures of the task. Our results show that this allocation is a central part of quantum model selection.

Tic-Tac-Toe exposes this point because its symmetries and decisive winning triples are known exactly. Full D_4 is the natural symmetry of the board, and enforcing it improves the standard edge ansatz over an unconstrained circuit. At the same time, the subgroup sweep shows that equivariance is not all-or-nothing: in our setting, C_4 sharing recovers almost the same generalization behavior as full D_4 . Thus, the strength of symmetry enforcement can itself be treated as an architectural

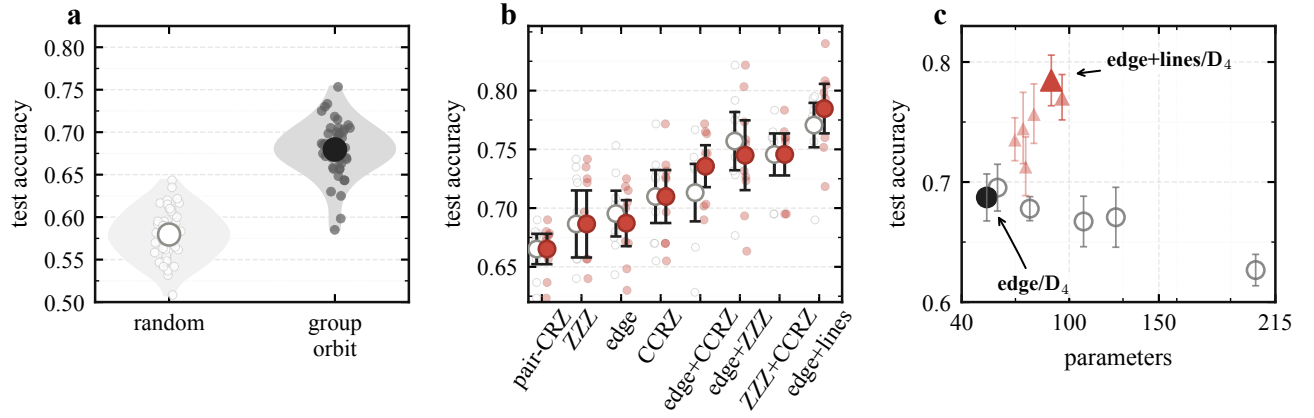


Fig. 3. Controls and ablations. (a) Parameter-matched random sharing underperforms group-orbit sharing. (b) Ablations are anchored by the labelled edge/ D_4 ansatz; the strongest model combines edge interactions with both line channels. (c) Accuracy is not explained by parameter count alone.

hyperparameter. Partial symmetry can preserve most of the useful bias while avoiding some of the rigidity introduced by the full group.

The larger effect comes from the choice of interactions. The edge ansatz respects neighboring board fields, but the label is generated by complete rows, columns, and diagonals. Adding orbit-shared ZZZ and CCR_Z gates on these winning triples keeps the model equivariant while placing trainable interactions directly on the motifs that define the task. The random-sharing, ablation, and parameter-count controls show that the improvement is not explained by fewer parameters or by adding gates indiscriminately. It comes from aligning the equivariant circuit with the geometry of the target function.

Our contribution is therefore a refinement of the symmetry program for variational quantum learning. Symmetry gives a principled way to restrict the hypothesis space; task structure determines how this restricted space should be used. A practical recipe is to represent the known data symmetry, choose the subgroup to enforce, and then place trainable interactions on label-relevant motifs. Tic-Tac-Toe is small and exactly enumerable, but precisely for this reason it exposes this principle without confounding factors. Future work should study softer versions of this recipe, such as regularized gate-specific deviations that interpolate between strict equivariance and fully free circuits, and apply motif-aligned equivariant ansätze to larger tasks where the relevant structures must be inferred from domain knowledge. In this sense, the operational question is not only whether to impose a symmetry, but how to use the remaining model capacity within the symmetry-respecting design space.

APPENDIX

A. Package

The [project repository](#) contains the material needed to reproduce the paper: experiment code, checked result tables, figure-generation scripts, and manuscript sources.

B. Data generation

Following Meyer et al. [8], we enumerate the 5478 legal, reachable boards, stop at terminal states, count unfinished positions as draws, and use class-balanced splits with a fixed 600-board test set. The stored tables record sampled indices, seeds, model family, subgroup, interactions, parameter count, and training size.

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